# Enhancing Requirement Analysis with Hybrid AI-Based Question Answering Systems

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**Abstract.** Requirement analysis is a critical phase in software engineering, but it is often affected by ambiguity, incompleteness, and inconsistency when Software Requirements Specifications (SRS) are written in natural language. Existing AI-based question answering systems improve requirement analysis but remain limited by the effectiveness of their retrieval and ranking components. This research enhances the QAssist framework by introducing BGE embeddings, a BGE cross-encoder reranker, and a hybrid retrieval strategy that combines lexical and semantic methods. The proposed approach is evaluated using the REQuestA dataset together with GTv6 and DexV2. Results demonstrate improved recall and ranking quality for domain-specific queries while introducing a trade-off in execution time. These findings highlight the importance of semantic retrieval in requirement analysis and demonstrate the potential of AI-based retrieval and question answering systems.

**Keywords:** Artificial Intelligence · Requirement Analysis · Question Answering · Information Retrieval · NLP

## 1 Introduction

Requirement analysis is a fundamental stage in software engineering that defines the expected functionality and behavior of a system. SRS are written in natural language to ensure accessibility for a range of stakeholders, including developers, managers, and domain experts. Natural language introduces several challenges, such as ambiguity, inconsistency, and incompleteness, which can significantly affect the quality of the final system [3], [4], [6]. These issues often lead to miscommunications, incorrect implementations, and increased development costs. The QAssist framework was introduced to address these challenges by applying question answering techniques to requirement analysis, allowing users to interact with requirements through natural language queries [1]. QAssist retrieves relevant information from both the SRS and external domain knowledge sources, where domain-specific corpora have been shown to improve the handling of ambiguity

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in requirements [5]. This approach highlights the potential of combining Natural Language Processing (NLP) and QA systems to support requirement engineers. Despite its effectiveness, the performance of such systems is mainly dependent on the retrieval and ranking components, which motivates further improvements.

### 1.1 Research Aim and Objectives

The main aim of this study is to enhance requirement analysis using AI-based retrieval and question answering systems. This involves improving the underlying retrieval and ranking mechanisms to better handle natural language queries. One objective is to replace existing dense retrieval models with more advanced embedding models that can capture semantic meaning more effectively. Another objective is to improve ranking performance using cross-encoder models that evaluate the relationship between queries and retrieved passages. The study also aims to introduce a hybrid retrieval approach that combines lexical and semantic methods, ensuring that both exact matches and contextual relevance are considered. Additionally, the research evaluates the effectiveness of these improvements using real datasets to ensure practical relevance. By achieving these objectives, the study contributes to the development of more robust QA systems for requirement analysis.

### 1.2 Research Problem

Although QA-based systems such as QAssist [1] provide a promising approach to requirement analysis, they still face limitations in accurately retrieving and classifying relevant information. One of the key challenges lies in handling domain-specific queries, which often require deeper semantic understanding beyond simple keyword matching. Traditional retrieval methods may fail to capture the context of such queries, leading to incomplete or irrelevant results [10]. Additionally, ranking mechanisms may not always prioritize the most relevant passages, which affects the overall usability of the system. These limitations highlight the need for improved retrieval and ranking techniques that can better handle the complexity of natural language requirements. Therefore, the objective of this research is to enhance the performance of QA-based requirement analysis systems by improving their ability to retrieve and rank relevant information. This is achieved by integrating advanced embedding models and hybrid retrieval strategies. By addressing these challenges, the study aims to provide a more effective tool for requirement engineers.

### 1.3 Research Questions

This research is guided by several important questions that aim to evaluate the effectiveness of the improvements.

**RQ1**: How do advanced embedding models influence retrieval performance, particularly in domain-specific contexts?

**RQ2**: Does a hybrid retrieval approach provide better results compared to traditional lexical or dense retrieval methods?

**RQ3**: What trade-offs exist between recall, ranking quality (nDCG), and execution time in the improved, more complex system?

### 1.4 Significance of the Study

This study shows how modern NLP and AI techniques can improve the process of requirement analysis [2]. By using automated querying and the retrieval of requirement information, the system reduces the manual effort required by analysts. It improves retrieval of relevant requirement information, which may assist analysts when identifying ambiguity and incompleteness [3], [4]. The introduction of hybrid retrieval shows the importance of combining multiple approaches to achieve better performance. The study shows trade-offs between retrieval accuracy, ranking quality, and computational efficiency. These insights are valuable for researchers and engineers working on AI-assisted requirement engineering. The findings contribute to the broader field of information retrieval by showing the effectiveness of hybrid methods in domain-specific contexts [13], [15]. This research shows the practical benefits of integrating AI into requirement analysis.

## 2 Proposed Methodology

### 2.1 Baseline

The baseline system used is based on the QAssist framework [1], which follows a standard open-domain question answering architecture. QAssist combines lexical and semantic retrieval methods to identify relevant information from SRS and domain-specific corpora. In the retrieval stage, the system uses BM25, a probabilistic information retrieval model [13] that ranks documents based on term frequency and inverse document frequency, making it effective for keyword-based matching [9]. In addition to BM25, QAssist employs DistilBERT for dense retrieval, which represents text as embeddings and captures semantic relationships between queries and passages [12]. This combination allows the system to retrieve candidate passages that are lexically and semantically relevant. After retrieval, the system applies a reranking step to refine the order of the retrieved passages, ensuring that the most relevant results are prioritized. Finally, a machine reading comprehension component extracts the answer from the top-ranked passages. While this pipeline is effective, it has limitations in capturing deeper semantic meaning, particularly for domain-specific queries where exact wording may differ.

### 2.2 Framework Overview

The proposed system follows a structured framework that extends the QAssist pipeline [1], [10]. The process begins with a user providing a natural language

query related to the requirements. The system then retrieves relevant documents from a domain-specific corpus using retrieval models. These documents are processed to extract smaller text passages that are more suitable for analysis. A hybrid retrieval step combines results from lexical and semantic methods to generate a set of candidate passages. These candidates are re-ranked using a cross-encoder model, which evaluates the relevance of each passage to the query. Finally, the system extracts the most relevant answer from the top-ranked passages.

### 2.3 Proposed Improvements

In this work, the QAssist framework [1] is enhanced by replacing its dense retrieval component with the BAAI/bge-large-en-v1.5 embedding model [7]. The ranking stage is further enhanced by incorporating a BGE cross-encoder for reranking [14], which jointly evaluates query–passage relevance before producing the final ranking. The original QAssist system [1] relies on DistilBERT for dense retrieval, which provides a reasonable balance between efficiency and performance, but it might not fully capture complex semantic relationships in domain-specific queries. By introducing BGE embeddings, which are optimized for semantic similarity tasks, the system is able to better understand contextual relationships between queries and requirement text [11]. This results in improved retrieval of relevant passages, particularly in cases where the wording of the query differs from the original requirement text. In addition, BGE embeddings provide stronger generalization capabilities across different domains, making them well-suited for requirement analysis tasks. This improves the baseline retrieval approach, because it directly enhances the system's ability to identify relevant information based on meaning rather than exact keyword matching.

Unlike retrieval models that independently encode queries and passages [12], the BGE cross-encoder processes both inputs jointly, allowing it to better evaluate their semantic relationship. This results in more accurate ranking of candidate passages, ensuring that the most relevant results are prioritized. The original QAssist evaluation employed a MiniLM-based cross-encoder for reranking [1], [8]. In this work, that component is replaced with a BGE cross-encoder reranker, which provides a strong balance between computational efficiency and ranking quality for semantic retrieval. The original QAssist system [1] uses simpler ranking mechanisms, while the improved approach significantly enhances the quality of the final ranked results. Furthermore, the improved ranking mechanism helps address cases where multiple passages are retrieved but differ in relevance, ensuring that the most useful information is presented to the user. As a result, the combination of BGE model [7] and BGE cross encoder [14] improves both retrieval accuracy and ranking precision.

### 2.4 Hybrid Retrieval Strategy

The hybrid retrieval approach is a key contribution. It combines the strengths of BM25 and dense retrieval methods to improve overall performance. Specifically,

BM25 [9] is used with BGE embeddings [7] to ensure that both exact keyword matches and semantic similarities are captured during retrieval. BM25 ensures that exact matches are not missed, while dense retrieval captures semantic similarities between queries and passages [9]. The ranked outputs from both methods are combined using Reciprocal Rank Fusion (RRF), where each candidate receives a fused score based on its ranking in each retrieval method before being passed to the reranking stage. This approach improves recall by increasing the likelihood of retrieving relevant information. It enhances the system's ability to handle domain-specific queries, which require semantic understanding and where relevant information may not share exact wording. These improvements increase recall but also introduce trade-offs in execution time, as the system becomes more computationally complex. The hybrid approach represents a practical solution to the limitations of individual retrieval methods. It is useful in requirement analysis, where lexical and contextual information are important. Overall, the combination of enhanced embeddings, improved reranking, and hybrid retrieval results in a more robust and effective QA system for requirement analysis, improving the original QAssist framework [1].

### 2.5 Role of NLP

NLP plays a central role by enabling the processing and understanding of textual data [2], [15]. It allows the system to break down requirements into tokens and sentences, which are analyzed more effectively. NLP techniques are used to convert text into embeddings [12], which represent the semantic meaning of words and phrases. These embeddings help performing similarity comparisons during retrieval. Additionally, NLP models are used in the extraction phase to identify relevant text spans within passages. This enables the system to provide precise answers to user queries. NLP serves as the foundation for the entire system.

### 2.6 Pipeline

The system follows a structured pipeline that processes user queries through multiple stages to generate accurate answers. The process begins with a natural language query provided by the user, which is the input. In the first stage, document retrieval is performed using both BM25 [9] and dense retrieval models to identify relevant documents from the domain-specific corpus [12]. These documents are divided into smaller segments during the passage extraction stage, allowing for more precise analysis of relevant information. Next, the results of lexical and semantic retrieval methods are fused using Reciprocal Rank Fusion (RRF) to generate a unified candidate set. The fused retrieval stage generates a candidate set of the top 10 passages. These candidates are then re-ranked using the BGE cross-encoder, which evaluates the semantic relevance of each passage with respect to the query [14]. Performance is subsequently evaluated at K = 1, 3, 5, and 10 by selecting the corresponding number of highest-ranked passages. After ranking, the top passages are selected and passed to the answer extraction stage, where the final answer is identified. This pipeline ensures that

both keyword-based and semantic information are utilized effectively throughout the retrieval process.

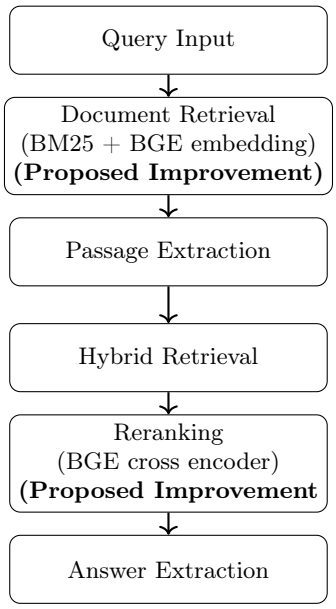



**Fig. 1.** Proposed QA pipeline for requirement analysis. Components marked **(Proposed Improvement)** indicate the two enhancements over the baseline QAssist system: (1) Document Retrieval replaces DistilBERT with BAAI/bge-large-en-v1.5 embeddings fused with BM25 via Reciprocal Rank Fusion, and (2) Reranking replaces the MiniLM cross-encoder with the BGE cross-encoder (BAAI/bge-reranker-v2-m3). All other stages remain unchanged from the original framework.

## 3 Case Study and Data Preparation

### 3.1 Dataset

The evaluation is based on the REQuestA dataset used in the QAssist study [1]. This dataset contains 387 question-answer pairs derived from multiple SRS documents across different domains. The questions are designed to test the system's ability to retrieve and understand requirement information. The dataset includes manually created and automatically generated questions. This diversity ensures that the system is tested under realistic conditions. The dataset includes domain-specific corpora, which provide additional context for answering queries [6]. These corpora are constructed using sources such as Wikipedia. Overall, REQuestA provides a comprehensive benchmark for evaluating QA systems.

### 3.2 Ground Truth

GTv6 is the ground truth dataset for evaluating the performance of the system. This dataset provides validated answers for each query, allowing for comparison between system generated results and expected outputs. The evaluation relies on the ground truth which ensures that performance is measured against a consistent benchmark. This allows for accurate performance measurement and reduces ambiguity in evaluation results. This dataset is essential for computing recall and nDCG as it allows the system to know whether relevant results have been retrieved and correctly ranked. GTv6 supports fair comparison between the baseline QAssist system and the improved pipeline by maintaining a standardized reference point across all experiments. By using a well defined ground truth dataset, the study ensures that the evaluation is reliable and reproducible.

### 3.3 Additional Dataset

DexV2 is an additional dataset to support the retrieval process and evaluate the system across different domains. DexV2 functions as a searchable corpus or index from which relevant passages are retrieved [6]. In the QAssist framework, this dataset is typically preprocessed collections of textual documents that have been structured to support efficient information retrieval. These documents are segmented into passages and indexed to allow fast matching with user queries. When a query is issued, the retrieval model searches in DexV2 to identify the most relevant passages, and then passed to the ranking and answer extraction. The dataset helps determine the quality of retrieved information, as it defines the pool of knowledge available to the system. By using DexV2, the evaluation is able to measure retrieval performance and generalization across different domains.

### 3.4 Data Preparation

The data used is preprocessed using a series of NLP techniques to ensure that it is suitable for retrieval and analysis. Initially, the textual content from the SRS and domain-specific corpora is cleaned to remove irrelevant symbols and inconsistencies. Then, the text is tokenized, which involves breaking down sentences into smaller units such as words or subwords to facilitate further processing. Following tokenization, the documents are divided into smaller passages through a process known as passage splitting, allowing the system to retrieve more precise and contextually relevant segments of text. Following the preprocessing strategy of the original QAssist framework, each paragraph was treated as a passage unless it exceeded 512 tokens, in which case it was split into overlapping passages to preserve contextual continuity. These passages are transformed into vector representations using embedding models, enabling efficient semantic retrieval based on similarity measures. In parallel, lexical indices are constructed using BM25 to support keyword-based retrieval [9]. Additionally, domain corpora are prepared to act as a searchable knowledge base during the retrieval phase. This combination of preprocessing steps ensures that both lexical and semantic information are effectively captured, improving the performance of the system.

## 4 Results and Discussion

The results demonstrate that the improved system consistently outperforms the baseline in retrieving and ranking domain-specific queries. The reported values represent averages across the evaluation dataset.

### 4.1 Experimental Setup

The experimental evaluation was performed using the REQuestA dataset together with the GTv6 ground truth and the DexV2 retrieval corpus. Performance was evaluated using Recall and nDCG for K = 1, 3, 5, and 10. The improved pipeline combines BM25 lexical retrieval with BGE semantic retrieval using Reciprocal Rank Fusion, followed by cross-encoder reranking before selecting the final ranked passages.

**Table 1.** Recall Comparison Across Top-K for Domain Queries

| K | Baseline Recall | Improved Recall | Recall Improvement (%) |
|---|---|---|---|
| 1 | 0.649 | 0.735 | +13.30 % |
| 3 | 0.813 | 0.841 | +3.44 % |
| 5 | 0.813 | 0.862 | +6.03 % |
| 10 | 0.813 | 0.862 | +6.03 % |

### 4.2 Retrieval Effectiveness Analysis using Recall

Table 1 presents the recall comparison between the baseline and improved systems across different K, which represent the number of top retrieved results considered during evaluation. The results are averaged across multiple SRS datasets, ensuring a consistent comparison between the two approaches. The improved system at K = 1 is where the recall shows the largest improvement. The improved system has a percentage improvement ranging from 3.44% to 13.30%, indicating consistent gains across K. These results demonstrate that the improved system retrieves more relevant results than the baseline at every retrieval depth [11]. The highest percentage improvement occurs at K = 1, indicating that the improvements are most impactful when retrieving a smaller number of top results. As K increases, the percentage improvement gradually decreases, suggesting that the relative benefit becomes smaller at higher retrieval depths. Overall, Table 1 shows a clear improvement in recall performance for the proposed system.

**Table 2.** nDCG Comparison Across Top-K for Domain Queries

| K | Baseline nDCG | Improved nDCG | nDCG Improvement (%) |
|---|---|---|---|
| 1 | 0.649 | 0.735 | +13.30 % |
| 3 | 0.753 | 0.803 | +6.64 % |
| 5 | 0.753 | 0.813 | +7.97 % |
| 10 | 0.753 | 0.813 | +7.97 % |

### 4.3 Ranking Quality Evaluation using nDCG

Table 2 presents the nDCG comparison between the baseline and improved systems across different K, providing insight into ranking quality. The improved system has a significant improvement in ranking performance at K = 1. The percentage improvement ranges from 6.64% to 13.30%, showing consistent gains across different values of K. These results indicate that the improved system not only retrieves relevant results but also ranks them more effectively [11]. The improvements in nDCG demonstrate that relevant information is positioned higher in the result list. The largest improvement occurs at K = 1, suggesting that the system is particularly effective at ranking the most relevant result at the top position. As K increases, the percentage improvement becomes more stable, indicating consistent ranking performance across larger result sets. Overall, Table 2 highlights the enhanced ranking performance of the improved system.

**Table 3.** Execution Time Comparison Across Top-K for Domain Queries

| K | Baseline Time (s) | Improved Time (s) | Time Change (s) |
|---|---|---|---|
| 1 | 2.312 | 8.691 | +6.379 |
| 3 | 2.312 | 8.691 | +6.379 |
| 5 | 2.312 | 8.691 | +6.379 |
| 10 | 2.312 | 8.691 | +6.379 |

### 4.4 Computational Cost and Execution Time Analysis

Table 3 presents the execution time comparison between the baseline and improved systems across K, providing a measure of system efficiency. The baseline has constant execution time of 2.312 seconds and the improved system consistently takes 8.691 seconds. This results in an increase of 6.379 seconds for every configuration. The consistent time across K suggests that the computational cost is influenced by the model complexity. Execution time was measured from the beginning of document retrieval until completion of the reranking process using the time.time() function from Python's standard library. The identical execution times across different values of K occur because retrieval, fusion, and reranking are performed once over the same candidate set, while changing K only affects the number of results reported during evaluation. The improved system requires

more processing time because of the advanced embedding model and reranking technique. Despite the increased time, retrieval and ranking performance remain consistent across all values of K.

### 4.5 Key Retrieval and Ranking Performance Findings

Examining performance across retrieval depths reveals additional distinctions between the baseline and improved systems. The results show that the improved system is more effective at retrieving relevant requirements even as the number of retrieved candidates increases. The baseline shows that the recall values stop improving significantly after K = 3. This suggests that the baseline approach is limited in its ability to discover additional relevant information beyond a certain point. The improved system continues to identify new relevant results at higher values of K, indicating that it operates over a deeper and more informative retrieval pool. In terms of nDCG, the improved system also achieves higher scores, indicating that relevant results are not only retrieved but also ranked more effectively. This means that important requirements are more likely to appear at the top of the result list.

### 4.6 Semantic Retrieval Analysis and Performance Interpretation

The results indicate that the proposed improvements enhance the system's ability to retrieve and rank relevant information, primarily by addressing what can be described as the semantic gap in requirement analysis. Specifically, the improved system uses BGE embeddings to capture semantic relationships, allowing it to link related domain concepts even when the wording differs. This is particularly important in requirement engineering, where domain-specific terms and implicit knowledge are common. The improvements in recall and nDCG suggest that the system is better able to understand the meaning of queries rather than relying solely on keywords. The increase in execution time reflects the additional complexity introduced by the improved models, particularly the use of BGE for reranking. However, this complexity contributes directly to better ranking quality, as shown by the higher nDCG values. These results demonstrate that the improved system provides a more semantically aware retrieval process.

Another aspect of the results is the relationship between retrieval depth and the effectiveness of semantic retrieval methods. The improved system demonstrates stronger gains at lower values of K, particularly at K=1, where the percentage improvement in recall and nDCG is the highest. This indicates that the system is more effective at identifying the most relevant result early in the ranking process. This reduces the need for users to manually search through multiple results, as the most useful information is likely to appear at the top of the list. As K increases, the performance gap between the baseline and improved systems decreases slightly, suggesting that both are able to retrieve a larger portion of relevant information at higher retrieval depths. However, the improved system consistently maintains higher recall and nDCG values, indicating that it is more effective overall. This behavior reflects the advantage of combining lexical

and semantic retrieval, as it allows the system to capture both exact matches and contextual relationships. The improved ranking performance suggests that the BGE cross-encoder is effectively prioritizing relevant passages within the candidates [14]. Overall, the results show that the proposed system not only improves retrieval performance but also enhances the quality and usability of the returned results.

### 4.7 Impact of Hybrid Retrieval on Retrieval Effectiveness

The hybrid retrieval approach introduces trade-offs [13]. The results show that recall improves significantly, while the nDCG results clarify that ranking quality also improves. Since nDCG values increase across K, the system is ranking relevant results higher. The primary challenge introduced by the hybrid method is not ranking accuracy, but increased computational complexity due to combining multiple retrieval signals. In some cases, hybrid retrieval may introduce subtle ranking inconsistencies, where a strong lexical match is ranked above a semantically richer passage, but this is not reflected as a dominant issue in the results. Overall, the improved system demonstrates that combining lexical and semantic retrieval can enhance performance while introducing feasible trade-offs.

The improvements can be explained through the concept of the semantic gap in requirement analysis. Traditional keyword-based retrieval methods, such as BM25, rely on exact term matching and may fail when different terminology is used to describe similar concepts [9]. For example, a requirement referring to system reliability may not be retrieved when a query uses the term “fault tolerance”, even though both refer to related concepts. In contrast, BGE embeddings are able to capture semantic relationships between these terms, allowing the system to link related domain concepts effectively. This ability to bridge the semantic gap is important in requirement engineering, where domain-specific terminology and implicit knowledge are common. As a result, the improved system is better able to retrieve relevant requirements even when exact keywords do not match.

### 4.8 Computational Trade-offs in Model Complexity

The increase in execution time from 2.312 to 8.691 seconds represents a significant computational cost. This increase is due to the use of more advanced embedding models and the BGE cross-encoder for reranking [14], both of which require additional processing time. In requirement engineering, this trade-off is considered acceptable, as accuracy is generally prioritized over real-time performance.

### 4.9 Effect of Proposed Enhancements on System Performance

The improvements indicate that the system is more robust when handling domain-specific queries, which involve complex terminology and implicit relationships. By combining lexical and semantic retrieval, the system can capture a wider range of relevant information. This leads to a comprehensive and accurate representation of the requirements.

### 4.10 Accuracy vs Efficiency Trade-off Analysis

The results highlight a trade-off between retrieval performance and computational efficiency. While the improved system achieves higher recall and nDCG across K, this improvement comes at the cost of increased execution time. Overall, the results confirm that improving retrieval performance and ranking quality introduces a trade-off with computational efficiency.

### 4.11 Effect of K

The results show that recall and nDCG both improve as the value of K increases for both the baseline and improved systems. Here, K represents the number of top retrieved results considered during evaluation, meaning that larger values of K allow the system to examine more candidate passages. For the baseline system, recall improves from 0.649 at K = 1 to 0.813 at K = 10, indicating that retrieving more results increases the likelihood of including relevant passages. However, the baseline system shows limited improvement beyond K = 3, suggesting that its retrieval capacity is constrained. In contrast, the improved system continues to improve at higher values of K, with recall reaching 0.862 and nDCG reaching 0.813. This indicates that the improved system is able to access a deeper and more relevant pool of candidate passages. The continued improvement at higher values of K suggests that the hybrid retrieval approach is more effective at identifying additional relevant information. The decreasing percentage improvement reflects the fact that both systems approach their maximum retrieval capacity as K increases. However, the improved system consistently maintains an advantage over the baseline. Overall, the results show the choice of K influences both retrieval performance and the relative effectiveness of the proposed improvements.

### 4.12 Evaluation of Research Questions Based on Experimental Results

The experimental results are interpreted with respect to the research questions defined in Section 1.3.

RQ1, which investigates the impact of advanced embedding models on retrieval performance, is addressed through the recall improvements shown in Table 1. The consistent increase in recall across all values of K demonstrates that BGE embeddings enhance the system's ability to retrieve relevant information, particularly for domain-specific queries where semantic understanding is required.

RQ2, which examines the effectiveness of hybrid retrieval, is supported by improvements in both recall and nDCG (Tables 1 and 2). The combined gains in retrieval depth and ranking quality indicate that integrating lexical and semantic methods provides more robust performance than relying on a single retrieval approach.

RQ3, which focuses on trade-offs between recall, ranking quality, and execution time, is addressed by comparing Tables 1–3. While the improved system

achieves higher recall and nDCG, Table 3 shows a significant increase in execution time. This confirms that enhanced performance comes at the cost of higher computational complexity. Overall, the results demonstrate that the proposed improvements successfully enhance retrieval effectiveness and ranking quality while introducing a trade-off in efficiency.

## 5 Limitations

Despite the improvements introduced in this study, several limitations remain that affect the overall effectiveness of the system. One major limitation is the dependency on dataset quality, as the performance of the QA system is directly influenced by the completeness and correctness of the underlying data. If the SRS or the domain-specific corpus contain missing or poorly structured information, the system may fail to retrieve accurate results. Another limitation is the increased computational cost associated with using advanced models such as BGE embedding model and BGE cross-encoder reranker, which require more processing time and resources compared to simpler baseline methods. This becomes particularly significant when scaling the system to larger datasets or real world industrial environments. Additionally, the system still struggles with highly ambiguous or vague requirements, where semantic retrieval methods may not fully capture the intended meaning. The hybrid retrieval approach improves recall but introduces challenges in ranking. Finally, the absence of user feedback mechanisms limits the system's ability to adapt or improve over time, which could otherwise enhance its practical usability. Additionally, this study evaluates the proposed retrieval pipeline as a complete system rather than performing an ablation study. The proposed system modifies three components simultaneously: the embedding model, the retrieval strategy, and the reranker. As a result, the observed gains in recall and nDCG cannot be attributed to any individual change. Therefore, it is not possible to determine whether the improvement is driven primarily by the stronger embeddings, the hybrid retrieval fusion, the improved reranker, or some combination of all three. Future work should isolate each component independently to establish which contributes most to the overall performance gain.

### 5.1 Threats to Validity

The validity concerns most relevant when interpreting the findings of this study are internal and external validity.

Internal Validity. The primary threat to internal validity concerns the implementation and configuration of the retrieval pipeline. The reported performance depends on the selected embedding model, reranking model, and retrieval parameters. To reduce this threat, the improved system was evaluated using the same benchmark datasets and evaluation metrics as the baseline, allowing the reported differences to reflect the proposed retrieval enhancements rather than changes in the evaluation benchmark.

External Validity. The evaluation was conducted using the REQuestA dataset together with the GTv6 ground truth and DexV2 retrieval corpus. Although these datasets provide a standardized benchmark for evaluating requirement analysis systems, the reported results may not generalize to all software engineering domains or industrial environments. Additional evaluation on larger and more diverse datasets would further strengthen the generalization of the findings.

## 6 Implications of the Research

The findings of this research extend beyond requirement engineering and have implications for multiple domains that rely on processing large volumes of unstructured text. For example, in legal document analysis, similar QA-based systems can be used to extract relevant clauses or identify inconsistencies across complex legal texts. In healthcare, the ability to retrieve and interpret domain-specific information from medical documents can assist clinicians in decision making and improve access to critical knowledge. Customer support systems can benefit from such approaches by enabling automated question answering over product documentation, reducing response times and improving user experience. The hybrid retrieval framework proposed highlights the importance of combining lexical and semantic methods, which is applicable to a wide range of information retrieval tasks. The observed trade-offs between recall and ranking quality provide valuable insights for designing systems that prioritize different performance aspects depending on the application. Overall, this research demonstrates that integrating AI with retrieval techniques can significantly enhance the processing and understanding of domain-specific information across various fields.

## 7 Future Work

There are several directions for future work that can improve the system [10]. One area is the integration of large generative models like ChatGPT which could enhance answer generation by providing more context aware and coherent responses. Another is improving the explainability of the system, allowing users to understand why certain answers are retrieved and how they relate to the requirements. This is important in requirement engineering, where traceability and justification are critical. Future work could focus on handling unanswerable queries more effectively by incorporating confidence scoring mechanisms or no answer detection models. Additionally, fine-tuning retrieval and ranking models on domain-specific datasets could improve performance, particularly in specialized industries where terminology and context vary. The hybrid retrieval framework could be optimized by balancing lexical and semantic components based on query type. Finally, conducting user studies with requirement engineers would provide valuable insights into the practical usability and effectiveness in real world scenarios.

## 8 Conclusion

This study demonstrates the potential of AI in enhancing requirement analysis through improved question answering systems [15]. By building upon the QAssist framework, this work introduces significant improvements in retrieval models and pipeline design, particularly through the use of hybrid retrieval methods. The results show that the improved system achieves higher recall for domain-specific queries, indicating a stronger ability to retrieve relevant information. However, this improvement comes with trade-offs, including increased computational cost and execution time, highlighting the complexity of balancing different performance metrics. The study emphasizes the importance of semantic understanding in processing natural language requirements and demonstrates how advanced NLP techniques can support analysts by improving retrieval of relevant requirement information. These findings contribute to the growing body of research on AI-assisted requirement engineering and provide insights into the design of more effective QA systems. Overall, this work shows that while improvements in one aspect of performance may introduce challenges in others, such trade-offs are essential considerations in the development of practical and scalable solutions.

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